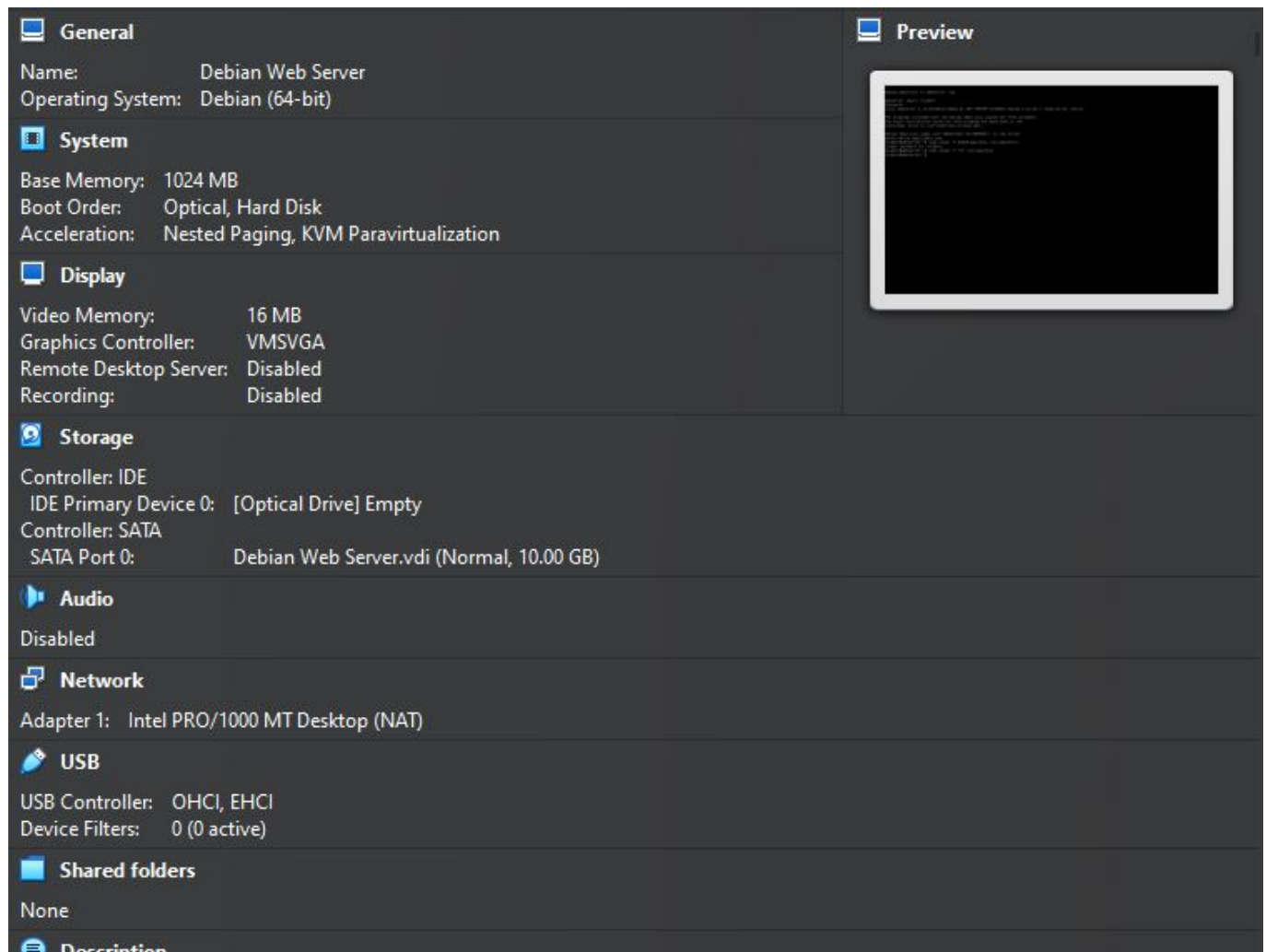
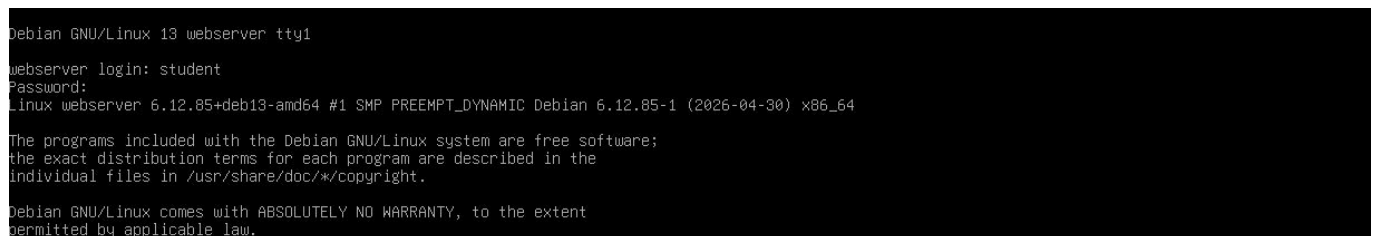


Deliverable 2

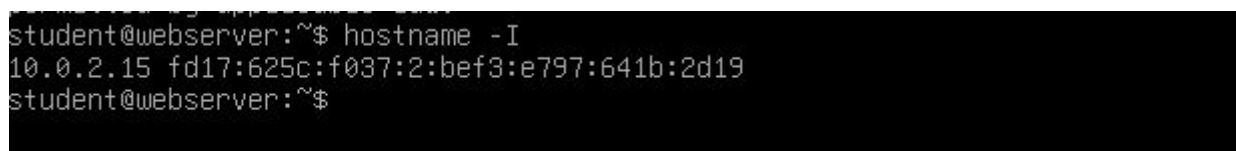
1. What are the server hardware specifications (virtual machine settings)?



2. What is the Debian Login Screen?



3. What is the IP address of my Debian Server Virtual Machine?



4. How do you work with the Firewall in Debian?

You can do things such as:

- `sudo ufw status` to tell you the status of the firewall
- `sudo ufw disable` to disable the firewall
- `sudo ufw allow 'Apache'` before you reenale the firewall `sudo ufw enable`

4a. How do you check if the Firewall is running?

```
student@webserver:~$ sudo ufw status
[sudo] password for student:
Status: active

To Action From
--
WWW Full ALLOW Anywhere
OpenSSH ALLOW Anywhere
WWW Full (v6) ALLOW Anywhere (v6)
OpenSSH (v6) ALLOW Anywhere (v6)
```

The `sudo ufw status` tells me the status of my firewall in the webserver VM.

4b. How do you disable the Firewall?

```
student@webserver:~$ sudo ufw disable
Firewall stopped and disabled on system startup
student@webserver:~$ _
```

`sudo ufw disable` to disable the firewall in the webserver VM.

4c. How do you add Apache to the Firewall?

```
student@webserver:~$ sudo ufw allow 'Apache'
Rules updated
Rules updated (v6)
```

`sudo ufw allow 'Apache'` allows you to add Apache into the firewall.

NOTE: Do this step above before you decide to enable the firewall again (`sudo ufw enable`)

5. What different commands do we use to work with Apache?

There are different commands we can use with Apache. For instance:

- `systemctl status apache2` will check the status of Apache in our webserver VM
- `sudo systemctl stop apache2` will halt Apache

5a. What is the command you use to check if Apache is running?

```
student@webserver:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-03 17:59:39 EDT; 16min ago
     Invocation: d0e9ea721a744fc6a6cedf1508a829fb
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 723 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 758 (apache2)
      Tasks: 55 (limit: 1110)
     Memory: 8.1M (peak: 8.3M)
        CPU: 154ms
    CGroup: /system.slice/apache2.service
            └─758 /usr/sbin/apache2 -k start
              └─759 /usr/sbin/apache2 -k start
                └─761 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
```

`systemctl status apache2` will check the status of Apache to see if it's working or not.

5b. What is the command you use to stop Apache?

```
student@webserver:~$ sudo systemctl stop apache2
```

`sudo systemctl stop apache2` will halt Apache.

5c. What is the command you use to restart Apache?

```
student@webserver:~$ sudo systemctl restart apache2
```

`sudo systemctl restart apache2` will restart Apache if it was off.

5d. What is the command used to test Apache configuration?

```
student@webserver:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress th
is message
Syntax OK
```

`sudo apache2ctl config test` will let you know if your Apache configurations are okay.

NOTE: `apache2ctl config test` does not work on it's own for me, so I used `sudo`.

5e. What is the command used to check the installed version of Apache?

```
student@webserver:~$ /usr/sbin/apache2 -v
Server version: Apache/2.4.66 (Debian)
Server built:   2026-03-01T13:26:45
```

`/usr/sbin/apache2 -v` will let you see what version of Apache you have installed.

NOTE: Running `apache2 -v` didn't work that well, so I decided to use the absolute directory.

5f. What are some common configuration files for Apache?

```
student@webserver:~$ ls /etc/apache2/
apache2.conf  conf-available  envvars  magic  mods-available  mods-enabled  ports.conf  sites-available  sites-enabled
```

`ls /etc/apache2/` will show you some of your config files for Apache.

NOTE: Be careful about `cat` into a file, some of them will run an insane block of information.

5g. Where does Apache store logs?

```
student@webserver:~$ sudo ls /var/log/apache2
[sudo] password for student:
access.log  error.log  other_vhosts_access.log
```

`sudo ls /var/log/apache2` will let you see where and what Apache stores its logs.

NOTE: I used `sudo` again because it will say you don't have permission to view the logs.

5h. What are some basic commands we can use to review logs?

```
student@webserver:~$ sudo head -n3 /var/log/apache2/access.log
::1 - - [02/May/2026:20:36:41 -0400] "GET / HTTP/1.1" 200 11014 "-" "Wget/1.25.0"
10.0.2.2 - - [03/May/2026:07:58:48 -0400] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
10.0.2.2 - - [03/May/2026:07:58:48 -0400] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://127.0.0.1:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
```

Some commands you can use to review logs are:

- `sudo head /var/log/apache2/access.log`
- `sudo tail /var/log/apache2/access.log`
- `sudo cat /var/log/apache2/access.log`

NOTE: This particular access.log will be massive, so you're best to use `head -n3`(for example) or `tail -n3` to view this file.